WSABI Summer Research Lab **Summer 2026**

Lecture 6: Statistical Inference

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6.1 Where We Are in the Course

In the first five lectures we built regression models: simple and multivariable linear regression, logistic regression, and GLMs/GAMs. We also emphasized a central warning: models do what they are told, so specification choices determine which patterns they can and cannot find.

This lecture asks a different question. Once a pattern appears, is it larger than what chance variation alone would plausibly produce?

Inference does *not* rescue a bad model or a confounded comparison. What it adds is a disciplined way to quantify how surprising the observed pattern would be under a stated null explanation.

For example:

- A basketball team shoots better after three days of rest. Is the rest effect real, or is this just game-to-game noise?
- A baseball player's batting average improves after a coaching change. Is this evidence that the coach helped, or is this regression to the mean?
- A diving judge gives higher scores to divers from his own country. Is that evidence of nationality bias, or could that happen by chance?
- A coefficient in a logistic regression model is positive. Is it large compared to the uncertainty in the estimate?

To understand a test statistic, we need to know what values we would expect it to take. A *null distribution* gives us exactly that.

> **Main idea.** Statistical significance is not about whether an observed effect is large in isolation.
> It is about whether the observed effect is large relative to what we would expect under the null.

6.2 Chance Variation

Sports data are noisy because athletes, teams, judges, and opponents vary in ways that are not fully explained by our measured variables. This unexplained variation is not always a mistake. It is often an unavoidable feature of the data-generating process.

Definition 6.1 (Chance Variation)**.** *Chance variation is the random-looking variation in observed data that remains even when there is no systematic effect of interest.*

A hard-hit ball can find a glove, a great shooter can miss an open three, and a judge can score a dive a bit high or low. That variability is part of the data-generating process, not automatically a model failure.

The question is not whether chance variation exists. It does. The question is:

> How large would an observed pattern need to be before chance variation no longer feels like a reasonable explanation?

This question is the starting point for significance testing.

6.3 The Four Ingredients of a Statistical Test

A statistical test requires four ingredients.

1. A **null hypothesis**, which formalizes the simple explanation.
2. A **test statistic**, which turns the data into a number measuring evidence against the null.
3. A **null distribution**, which says what values of the test statistic are expected under the null.
4. A **p-value**, which compares the observed test statistic to the null distribution.

Definition 6.2 (Null Hypothesis). *The null hypothesis, denoted H_0, is the hypothesis we temporarily assume to be true in order to measure how surprising the observed data are under a simple baseline explanation.*

Definition 6.3 (Research Hypothesis). *The research hypothesis, often denoted H_A or H_1, is the claim we are looking for evidence in favor of. It is usually the scientific or sports question that motivated the analysis.*

The null is a benchmark for chance variation.

Definition 6.4 (Test Statistic). *A test statistic is a function of the data,*

$$T = T(data),$$

chosen so that larger, smaller, or more extreme values represent stronger evidence against the null hypothesis.

For example, if we are studying whether a judge favors divers from his own country, a natural statistic is the difference between the judge's average discrepancy for same-country divers and his average discrepancy for other divers.

Definition 6.5 (Null Distribution). *The null distribution is the distribution of the test statistic T when the null hypothesis is true.*

The null distribution is the central object in this lecture. Without it, we cannot say whether an observed statistic is surprising.

6.4 Exact Null Distributions: Coin Flipping

Sometimes the null distribution is exact and easy to write down.

Example 6.6 (A Fair Coin)**.** Suppose we flip a coin n times and observe X heads. We want to test whether the coin is biased toward heads.

The null hypothesis is

$$H_0 : p = \frac{1}{2},$$

where p is the probability of heads.

A natural test statistic is

$$T = X = \text{number of heads.}$$

Under the null hypothesis, every flip is a Bernoulli random variable with probability 1/2 of heads, so

$$T \sim \text{Binomial}\left(n, \frac{1}{2}\right).$$

Therefore the null distribution is known exactly:

$$\mathbb{P}_{H_0}(T = k) = \binom{n}{k}\left(\frac{1}{2}\right)^n, \quad k = 0, 1, \ldots, n.$$

If we flip the coin 20 times and observe 16 heads, then the one-sided p-value for bias toward heads is

$$\begin{aligned} p\text{-value} &= \mathbb{P}_{H_0}(T \geq 16) \\ &= \sum_{k=16}^{20} \binom{20}{k}\left(\frac{1}{2}\right)^{20} \\ &\approx 0.0059. \end{aligned}$$

This calculation is possible because the null distribution is explicit. We know what chance variation looks like if the coin is fair.

6.4.1 One-sided and Two-sided Alternatives

If the research hypothesis is that the coin is biased *toward heads*, then large values of T are evidence against the null. The p-value is

$$\mathbb{P}_{H_0}(T \geq t_{\text{obs}}).$$

If the research hypothesis is that the coin is biased in *either direction*, then both unusually large and unusually small values of T are evidence against the null. A natural two-sided p-value is

$$\mathbb{P}_{H_0}\left(\left|T - \frac{n}{2}\right| \geq \left|t_{\text{obs}} - \frac{n}{2}\right|\right).$$

For $n = 20$ and $t_{\text{obs}} = 16$, the two-sided p-value is approximately 0.0118.

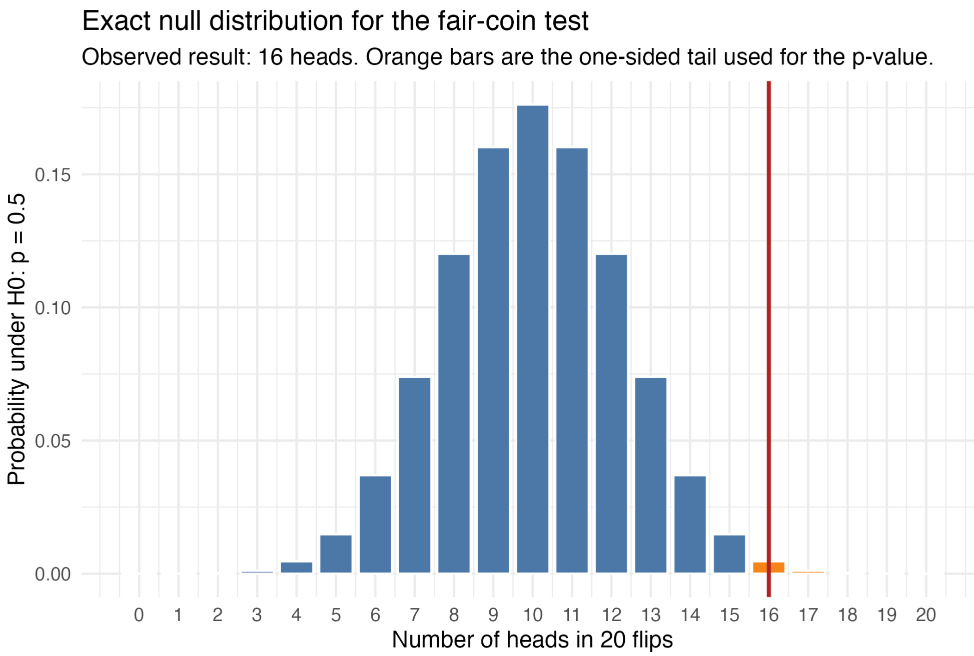

Figure 6.1: Exact null distribution for the number of heads in 20 flips of a fair coin. The red line marks the observed result, 16 heads, and the orange bars show the one-sided tail used for the p-value.

6.5 p-values

Definition 6.7 (p-value)**.** *A p-value is the probability, assuming the null hypothesis is true, of obtaining a test statistic at least as extreme as the one observed.*

For a one-sided test where larger values of T are more extreme,

$$p\text{-value} = \mathbb{P}_{H_0}(T \geq t_{\text{obs}}). \tag{6.1}$$

For a two-sided test where deviations in either direction are important,

$$p\text{-value} = \mathbb{P}_{H_0}(|T| \geq |t_{\text{obs}}|), \tag{6.2}$$

provided $T = 0$ represents the null value. The exact form depends on the statistic, but the idea is the same: compare the observed result to what the null says is typical.

> **Interpretation.** A small p-value means that the observed statistic would be unusual if the null hypothesis were true. It does not, by itself, prove that the research hypothesis is true.

Common misinterpretations:

- The p-value is not the probability that the null hypothesis is true.
- The p-value is not the probability that the result happened by chance.

- The p-value is not the size of the effect.
- A statistically significant result is not automatically a practically important result.
- A statistically significant association is not automatically a causal effect.

6.6 Statistical Significance

Definition 6.8 (Statistical Significance)**.** *An observed result is statistically significant at level α if its p-value is at most α.*

For example, at the $\alpha = 0.05$ level, a result is statistically significant if

$$p\text{-value} \leq 0.05.$$

The number 0.05 is a convention, not a law of nature. In practice, it is usually better to report the effect size, an uncertainty measure, and the p-value than to reduce the analysis to "significant" or "not significant."

6.7 A Model-Based Null Distribution: the Normality Assumption

The coin-flipping example is clean because the null distribution is exact. In many sports problems, we do not have such a simple exact null distribution.

Suppose $Y_1, \ldots, Y_n$ are measurements of performance under some condition, and we want to test whether the mean equals some null value μ_0.

A model-based approach might assume

$$Y_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2).$$

Under the null hypothesis $H_0 : \mu = \mu_0$, if σ is known, the statistic

$$Z = \frac{\bar{Y} - \mu_0}{\sigma/\sqrt{n}}$$

has the standard normal null distribution,

$$Z \sim \mathcal{N}(0, 1).$$

This gives a p-value. For a two-sided test,

$$p\text{-value} = \mathbb{P}_{H_0}(|Z| \geq |z_{\text{obs}}|).$$

Remark 6.9 (Normality in this lecture)**.** We have not yet derived the Central Limit Theorem. So here normality should be read as a modeling assumption or approximation, not as something already proved from first principles.

The important point is that every null distribution comes from assumptions: an exact probability model, a parametric model, or a randomization or exchangeability argument.

6.8 Regression Output and Null Distributions

Regression models have null distributions too, even if software hides the construction.

A regression p-value is not magic. It comes from a statement like this:

> If the null hypothesis $H_0 : \beta_j = 0$ were true, what values of the estimated coefficient $\hat{\beta}_j$ would we expect to see just from chance variation?

6.8.1 Two Layers of a Regression Model

Regression models usually make two related claims.

1. A claim about the **outcome distribution** conditional on predictors.
2. A claim about the **sampling distribution** of coefficient estimates or test statistics.

For example, in linear regression we often write

$$Y_i \mid X_i = x_i \sim \mathcal{N}(x_i^T \beta, \sigma^2).$$

This is a distributional assumption about the outcome Y_i given predictors X_i.

From this assumption we get a distribution for coefficient estimates. In the classical Gaussian linear model,

$$\hat{\beta}_j \approx \mathcal{N}\left(\beta_j, \operatorname{se}(\hat{\beta}_j)^2\right),$$

and, under $H_0 : \beta_j = 0$,

$$\frac{\hat{\beta}_j}{\operatorname{se}(\hat{\beta}_j)}$$

has a known or approximately known null distribution.

Remark 6.10 (Parameters versus estimators)**.** In frequentist inference, β_j is fixed and unknown, while $\hat{\beta}_j$ is random. So saying "the beta is approximately normal" is shorthand for saying that the estimator has an approximately normal sampling distribution.

6.8.2 Examples Across Models

Model	Outcome assumption	Mean model	Coefficient inference
Linear regression	$Y_i \mid x_i \sim \mathcal{N}(\mu_i, \sigma^2)$	$\mu_i = x_i^T \beta$	$\hat{\beta}_j$ normal/t
Logistic regression	$Y_i \mid x_i \sim \text{Bernoulli}(p_i)$	$\text{logit}(p_i) = x_i^T \beta$	$\hat{\beta}_j$ approx. normal
Poisson regression	$Y_i \mid x_i \sim \text{Poisson}(\lambda_i)$	$\log(\lambda_i) = x_i^T \beta$	$\hat{\beta}_j$ approx. normal
GAMs	Depends on outcome	$g(\mu_i) = \eta_i$ with smooth terms	Approximate tests for terms

Table 6.1: Regression models specify a distribution for the outcome and imply a null distribution for coefficient estimates or test statistics.

For logistic and Poisson regression, the normality of $\hat{\beta}$ is usually a large-sample approximation. Since we have not developed that theory yet, the right interpretation for now is:

> Software p-values in GLMs rely on model-based approximations to the null distribution of a coefficient estimate or related statistic.

This connects directly to model misspecification. If the outcome distribution, independence assumptions, variance assumptions, or functional form are badly wrong, then the null distribution used by the software may be wrong too. In that case, the p-value may no longer mean what it claims to mean.

6.9 Permutation Tests: Building a Null Distribution from Data

Now consider a setting where a clean parametric null distribution is hard to justify. This is common in sports analytics. Diving scores, referee calls, subjective ratings, and play-by-play outcomes often contain complicated dependence and non-normality.

Permutation testing gives us another way to construct a null distribution.

6.9.1 The Core Idea

A permutation test is appropriate when the null hypothesis implies that certain labels are exchangeable.

Definition 6.11 (Exchangeability for a Permutation Test)**.** *Labels are exchangeable under the null if, assuming the null hypothesis is true, the observed outcomes would have been equally plausible under many rearrangements of the labels.*

For example, suppose we have n observations, each with an outcome Y_i and a group label $G_i \in \{0, 1\}$. We want to test whether the group labels are associated with the outcomes.

A natural statistic is the difference in sample means:

$$T_{\text{obs}} = \bar{Y}_{G=1} - \bar{Y}_{G=0}. \tag{6.3}$$

Under the null hypothesis of no relationship between G and Y, the labels G_i can be permuted among the outcomes Y_i. Each permutation gives a new test statistic:

$$T^{(b)} = \bar{Y}^{(b)}_{G=1} - \bar{Y}^{(b)}_{G=0}, \quad b = 1, \dots, B.$$

The empirical distribution of $T^{(1)}, \dots, T^{(B)}$ is the permutation null distribution.

For a one-sided alternative where large values are evidence against the null, the Monte Carlo permutation p-value is often computed as

$$\widehat{p} = \frac{1 + \sum_{b=1}^{B} \mathbb{1}\{T^{(b)} \geq T_{\text{obs}}\}}{B + 1}. \tag{6.4}$$

The $+1$ correction prevents reporting a p-value of exactly zero when only finitely many random permutations are used.

6.9.2 Permutation Tests as a Response to the Null Distribution Problem

The permutation test does not remove assumptions; it replaces a distributional assumption with an exchangeability assumption. That is often easier to justify in randomized settings than in observational sports data.

> **Permutation logic.** If the null hypothesis says the labels do not matter, then randomly rearranging the labels should produce datasets that are plausible under the null.

6.10 Case Study: Nationality Bias in Diving Judges

We now apply these ideas to a diving example. The research question is whether judges systematically give higher scores to divers from their own country.

6.10.1 Discrepancies

For each judged dive, define a judge's discrepancy as

$$D_i = \text{judge's score on dive } i - \text{baseline score for dive } i.$$

The baseline could be the average score for that dive across judges or another measure of dive quality. The point is to compare judges after adjusting for the fact that some dives are simply better than others. Without this adjustment, a judge could look biased just because the nationality-matched divers happened to perform stronger dives.

For a given judge, define

$$M_i = \begin{cases} 1, & \text{if the diver's nationality matches the judge's nationality,} \\ 0, & \text{otherwise.} \end{cases}$$

The test statistic is the difference of discrepancies:

$$\text{DoD} = \bar{D}_{M=1} - \bar{D}_{M=0}. \tag{6.5}$$

A positive DoD means that the judge gives higher discrepancies, on average, to divers from the judge's own country than to other divers.

6.10.2 Why This Statistic Matches the Question

The DoD statistic asks:

> Among the dives this judge saw, are the nationality-matched dives disproportionately the ones with positive discrepancies?

That is much closer to the substantive question of nationality bias than simply comparing raw average scores across countries.

6.10.3 Hypotheses

The research hypothesis is:

$$H_A : \text{the judge is biased in favor of divers from the judge's own country.}$$

The null hypothesis is:

$$H_0 : \text{the judge's discrepancies are not systematically related to nationality matches.}$$

Under the null, once discrepancies have been computed, the nationality-match labels should be exchangeable with respect to the discrepancy values.

6.10.4 McFarland Example

For Judge McFarland, the observed data can be summarized as follows:

- He judged 657 dives.
- 42 of those dives involved American divers, matching his nationality.
- His average discrepancy for matched dives was about 0.20.
- His average discrepancy for non-matched dives was about 0.01.
- The observed DoD was about 0.19.

So after adjusting for dive quality, the nationality-match gap is about 0.19. The inferential question is whether a gap that large would be unusual if the 42 matched dives were effectively random among the 657 dives he judged.

6.10.5 Permutation Procedure

For Judge McFarland, the permutation test proceeds as follows.

1. Keep McFarland's 657 discrepancy values fixed.
2. Randomly choose 42 of the 657 dives to be labeled "matched." Label the remaining 615 dives "non-matched."
3. Compute the DoD for this randomized labeling.
4. Repeat many times to form a permutation null distribution.
5. Compare the observed DoD, 0.19, to this permutation null distribution.

Step 2 is the key step: it keeps the number of matched dives fixed while breaking any systematic link between nationality and scoring.

In the observed analysis, only about 1 out of 1000 permuted assignments produced a DoD as large as McFarland's actual value. Equivalently, the permutation p-value was around 0.001.

> This means that if nationality-match labels were irrelevant to McFarland's discrepancies, a DoD as large as the one observed would be very unusual under the permutation null distribution.

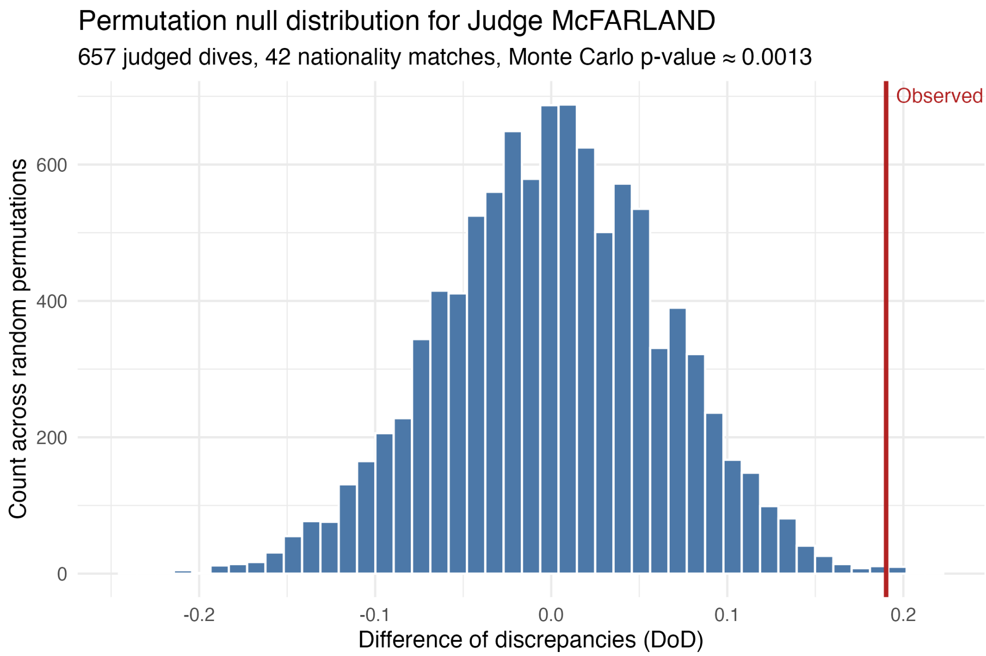

Figure 6.2: Permutation null distribution for McFarland's difference of discrepancies. The observed DoD is far out in the right tail, which is why the permutation p-value is very small.

This is statistical evidence against the null hypothesis of no nationality-related discrepancy. It is not, by itself, a complete explanation of why the pattern exists. It also depends on the validity of the discrepancy definition and the exchangeability assumption.

6.10.6 A Small Sample Can Still Be Informative

Small samples are weak when the data are ordinary, but they can still be informative when the pattern is extreme.

Suppose a judge has only 3 nationality-matched dives, but all three discrepancies are in the top 10% of that judge's discrepancy distribution. A rough approximation to the p-value is

$$(0.10)^3 = 0.001.$$

The exact permutation calculation would sample without replacement, so it is not exactly the same as multiplying independent probabilities. But the approximation explains the intuition: seeing all three matched dives among the most favorable discrepancies would be unusual under random labeling.

6.11 What Assumptions Buy Us

The theme of this lecture is that inference requires a null distribution. Different methods get that null distribution in different ways.

Method	Source of null distribution	Main assumption
Coin-flip test	Exact binomial distribution	Independent fair flips under H_0
Normal mean test	Normal model	Gaussian outcomes/errors
Regression coefficient test	Regression/GLM model	Correct outcome and error model, or good approximation
Permutation test	Re-labeling the data	Exchangeability under H_0

Table 6.2: Every test needs a null distribution, and every null distribution comes from assumptions.

Assumptions are what generate the null distribution, so we need to know exactly which ones we are using.

A useful statistical habit is to ask:

1. What is the test statistic?
2. What is the null hypothesis?
3. What is the null distribution?
4. What assumptions produce that null distribution?
5. Are those assumptions plausible in this sports context?

6.12 Significance versus Practical Importance

Statistical significance answers one question:

> Is the observed statistic surprising under the null distribution?

It does not answer:

> Is the effect large enough to matter for decision-making?

In sports analytics, this distinction is critical. A tiny effect can be statistically significant in a massive dataset. A large effect can fail to be statistically significant in a small dataset. Neither statement alone tells a coach, front office, or player what to do.

For any analysis, report both:

- an **effect size**, such as a difference in points, win probability, goals, expected points, or log-odds;
- an **uncertainty measure**, such as a standard error, confidence interval, or p-value.

6.13 Multiple Testing

The diving example naturally leads to a multiple-testing problem. If we test many judges, some small p-values can occur by chance even if every null hypothesis is true.

If we run 100 valid tests at the 0.05 level, and all null hypotheses are true, then the expected number of statistically significant results is

$$100 \times 0.05 = 5.$$

So if we screen many judges, a few small p-values are not surprising by themselves. Two common error targets are:

- **FWER (family-wise error rate):**
$$\mathrm{FWER} = \mathbb{P}(V \geq 1),$$
the probability of making *at least one* false discovery in the whole family of tests.
- **FDR (false discovery rate):**
$$\mathrm{FDR} = \mathbb{E}\left[\frac{V}{\max(R, 1)}\right],$$
the expected fraction of reported discoveries that are false.

Here V is the number of false discoveries and R is the total number of rejections.

- If falsely accusing even one judge would be a major problem, control **FWER**. The simplest tool is **Bonferroni**.
- If the goal is exploratory screening and we can tolerate some false leads, control **FDR**. The standard tool is **Benjamini–Hochberg**.

When many tests are performed, it also helps to:

- pre-specifying the main test before looking at the data;
- adjusting for multiple comparisons, using a target such as FWER or FDR;
- reporting all tested hypotheses, not only the significant ones;
- treating surprising results as exploratory unless confirmed on new data.

The p-value for one judge answers a question about that judge and that statistic. It does not automatically account for the fact that we may have searched across many judges, sports, or outcomes.

6.14 Significance Is Not Causality

A small p-value can reject a chance-based null without establishing causality.

Another way to say this is:

> Significance can rule out one particular chance-based null explanation. It does not rule out confounding, selection, reverse causality, or model misspecification.

For example, suppose a regression model finds that NBA players score more points in games where they play more minutes, and the coefficient is highly statistically significant. That does not mean that simply giving any player more minutes would cause that player to score at the same rate. Better players are selected to play more minutes.

Similarly, suppose a baseball model finds that players who changed coaches improved more than players who did not. That may be a coaching effect, but it may also reflect selection, regression to the mean, injury recovery, or other confounding variables.

The next conceptual step is causal inference, where we ask:

> When can an association be interpreted as a cause-and-effect relationship?

Causal inference requires both:

1. **Identification**: are we estimating the causal quantity we care about?
2. **Inference**: how uncertain is our estimate?

This lecture focused on inference. Causal inference adds identification.

6.15 Summary

The main lessons are:

- Chance variation is unavoidable in sports data.
- A test statistic is only interpretable after we know its null distribution.
- A p-value is the probability, under the null, of a result at least as extreme as the one observed.
- Exact null distributions exist in simple cases like coin flipping.
- Model-based null distributions rely on assumptions such as normality, Bernoulli outcomes, Poisson outcomes, independence, and correct functional form.
- Regression p-values come from null distributions for coefficient estimates or test statistics.
- Permutation tests build a null distribution by re-labeling data under an exchangeability assumption.
- Statistical significance is not the same as practical importance.
- Statistical significance is not the same as causality.

6.16 Exercises

1. A coin is flipped 30 times and lands heads 22 times. Write the null hypothesis, test statistic, null distribution, and one-sided p-value for testing whether the coin is biased toward heads.
2. In a basketball dataset, a team wins 18 of 25 games after two or more days of rest. Suppose the team's baseline win probability is 0.50. Under the null model that rest has no effect and each game is independent, what is the null distribution for the number of wins? What is the one-sided p-value?

3. In the diving example, explain why comparing a judge's raw average score for American divers to his raw average score for non-American divers is not enough to establish nationality bias.
4. Define the difference of discrepancies statistic for a judge. Why is it a better statistic than raw average score difference?
5. In your own words, explain the exchangeability assumption in the permutation test for diving judges.
6. Suppose a regression coefficient has estimate $\hat{\beta} = 0.12$ and standard error 0.04. Using a standard normal approximation, compute the two-sided p-value for testing $H_0 : \beta = 0$. What does this p-value mean?
7. Give an example of a statistically significant sports result that might not be practically important.
8. Give an example of a statistically significant association in sports that should not automatically be interpreted causally.